Artificial Intelligence

BS (CS) Spring 2026

Lab 05 Tasks

Learning Objectives:

1. Informed Search
2. A*Algorithm
3. Greedy Best-First Search

Lab Tasks

Submission Instructions

1. Create one notebook file for all questions with name *ROLLNO_SEC_LAB5* **e.g. i23XXXX_A_LAB05**
2. Submit your .ipynb file with correct name on Google Classroom under your section heading.
3. If you don't follow the above-mentioned submission instruction, you will be marked **zero.**

Plagiarism in the Lab Task will result in **zero** marks in the whole category.

Questions

Q1. A factory has a robot tasked with collecting parts from a workshop and delivering them to an assembly station. The workshop floor is represented as a **5×5 grid**, where each cell may contain a part to be collected. The robot starts at the top-left corner `(0, 0)` and must reach the delivery station at the bottom-right corner `(4, 4)`.

The robot can move in four directions: **Up, Down, Left, Right**, and each move has an associated traversal cost that may vary depending on obstacles, floor conditions, or other factors. The goal is for the robot to **collect the required parts and reach the destination efficiently**, prioritizing paths that appear closest to the goal.
Implement a **Greedy Best-First Search** algorithm to guide the robot from the start position to the destination while collecting parts. Use a **heuristic function** based on the **Manhattan distance** from the current position to the destination to determine which grid cell the robot should explore next.

Input:

1. **Grid:** A 5×5 grid representing the workshop floor, where each cell contains:
   - `0` for empty space
   - `1` for a cell containing a part
2. **Traversal Cost:** The cost of moving between adjacent cells (can vary per move).
3. **Start Position:** `(0, 0)`
4. **Destination:** `(4, 4)`

Heuristic Function:
Use the Manhattan distance as a heuristic:

$$h(n) = | x_{current} - x_{goal} | + | y_{current} - y_{goal} |$$

Output:

- The **sequence of grid positions** the robot moves through.
- The **order in which parts are collected**.
- The **total traversal cost** along the chosen path.

Requirements:

1. **Implement Greedy Best-First Search,** always expanding the node with the lowest heuristic value.
2. Avoid revisiting nodes unnecessarily to prevent cycles.
3. Display the **path**, the **collected parts**, and the **total cost**.

R	0	0	1	0
0	0	1	0	0
0	1	0	0	0
0	0	0	1	0
0	0	0	0	D

```
Path: (0,0) → (0,1) → (0,2) → (0,3) → (1,3) → (1,4) → (2,4) → (3,4) →
(3,3)  → (4,4)
Collected Parts at: (0,3), (3,3)
```

One possible solution is this.

=